

Appendix C: Interview Questions

State of Practice of {domain} Software

Information about these interview questions: This gives you an idea what I would like to learn about the development of {domain} software. Interviews will be one-to-one and will be open-ended (not just “yes or no” answers). Because of this, the exact wording may change a little. Sometimes I will use other short questions to make sure I understand what you told me or if I need more information when we are talking such as: “*So, you are saying that ...?*”, to get more information (“*Please tell me more?*”), or to learn what you think or feel about something (“*Why do you think that is ...?*”).

Questions about Interviewee

1. Interviewees’ current position/title? degrees?
2. Interviewees’ contribution to/relationship with the software?
3. Length of time the interviewee has been involved with this software?

Questions about Developers and Development Process, Tools and Techniques

4. How large is the development group?
5. What is the typical background of a developer?
6. What is your software development model? For example, waterfall, agile, etc.
7. What is your project management process? How is git used? Are there other project management tools?
8. Currently, what are the most significant pain points in your development process? Specifically, what are the pain points in the following stages of the development process: a) requirements definition (defining what the software should do), b) design, c) implementation, and d) verification.
9. How does documentation fit into your development process? Would improved documentation help with the obstacles you typically face?
10. In the past, is there any major obstacle to your development process that has been solved? How did you solve it?
11. Is it hard to ensure the correctness of the software? If there were any obstacles, what methods have been considered or practiced to improve the situation? If practiced, did it work?
12. When designing the software, did you consider the ease of future changes? For example, will it be hard to change the structure of the system, modules or code blocks? What measures have been taken to ensure the ease of future changes and maintenance?
13. Do you have any concern that your computational results won’t be reproducible in the future? Reproducibility is defined as the ability to obtain the exact same computational results in the future using the original dataset, code, software environment, and analytical steps. Have you taken any steps to ensure reproducibility?
14. What positive contributions do LLM-based tools have on the development process?
15. What negatives are associated with LLM-based tools?

Questions about Users

16. What is the typical background of a user?
17. Provide instances where users have misunderstood or misused the software. What, if any, actions were taken to address usability issues?
18. Do you think the current documentation can clearly convey all necessary knowledge to the users? If yes, how did you successfully achieve it? If no, what improvements are needed?
19. How do you learn about and understand the needs of your users? What channels do you use to gather user feedback?

END